Anish Natekar

Indian Institute of Technology, Goa

Fourth Year **Undergraduate**, **Computer Science and Engineering**
Address: B-701, Silver Oak, Ridge St., Hiranandani Gardens, Powai, Mumbai - 400076
E-mail: **anish.natekar.20031@iitgoa.ac.in**
Mobile: **+91 9833006472**
LinkedIn: **www.linkedin.com/in**
Github: **https://github.com/anish-natekar**

Education

BTech, Computer Science and Engineering, Indian Institute of Technology Goa	CGPA : **8.35/10**	2020 – 2024
Class 12, ISC (ICSE), Hiranandani Foundation School, Powai	Aggregate: **86.8** %	2018 – 2020
Class 10, ICSE, Hiranandani Foundation School, Powai	Aggregate: **94** %	2018

Experience

Researcher at TCS Research working in Generative AI for Media Research Area (August 2024 – Present)

- Working on Generative AI research projects for Multi Media applications.
- Working with LLMs, multi agent systems, Transformers, Diffusion models, Reinforcement Learning, Prompt Engineering, etc to help solve complex creative real world tasks.

BTech Project, HyP-ECA: An Attention for Aerial Tree Crown Deliniation and Species classification (Accepted, CVIP 2024) (Jan 2024 – May 2024)

[Authors: Anish Natekar (FIRST AUTHOR), Vipin Gautam, Prahalad Vijaykumar, Guidance of Dr. Shitala Prasad Dr. Clint P George | IIT Goa]

- Worked on creating a complete pipeline for Aerial view Forest survey for Tree Species Detection in Forest using GPS autonomous Quadcopter.
- Curated and Hand Annotated a Muti Tree Species Aerial Detection Dataset (MTAD) from various Open Source Datasets of single Tree Species.
- Trained a customized YOLO V8 Object Detector with our Novel Hyp-ECA Attention Mechanism for detecting smaller Tree Species better.
- Made a Pixhawk+Raspberry Pi GPS Autunomous Quadcopter UAV for performing Aerial Survey. Used Web-ODM software for photogrametry.
- Developped a ROS+Gazebo+PX4 autopilot+QGC SITL Simulation for Testing UAV Survey Scripts in Virtual 3D world called "Baylands.world".

Research and Development Intern Software at Siemens EDA (previously Mentor Graphics) (July 2023 – Dec 2023)

- Worked on the compiler optimization stage (vopt) in Questa sim a Hardware Description Language compiler/simulator by Mentor.
- Worked on X-value propagation in optimization stage (vopt). Made enhancements, regression test cases, and bug fixes in the Questa codebase.
- Codebase was in C on a Centos Linux server, vim editor, and version controller Perforce Helix.

RnD Intern, Efficient Parallel Simulation of Networked Synchronous Discrete-Event Systems (Accepted, Winter Sim 2024) (June 2022 – Aug 2022)

[Guidance of Dr. Neha Karanjkar | IIT Goa]

- Performed tests, simulations and suggested features for Sitar V2, developed by my supervisor herself during her Masters and PhD at IIT Bombay.
- Updated the framework that was tested on Ubuntu 12.04 LTS to make it compatible with Ubuntu 20.04 LTS and Python 3.
- Read many papers and wrote the literature review and performance analysis section of the HiPC paper using LATEX in IEEE format.
- Wrote PyPy optimized Simpy models to compare sequential performance of O3 gcc optimized Sitar and SystemC models.
- Built the ReadTheDocs documentation site for Sitar V2 along with my teammate to help new users learn about Sitar, use and contribute to it.

Projects

Created Raspberry Pi PicoW Copter a WiFi based Open Source affordable 60g Quadcopter (50+ stars on GitHub) [Github] (May 2023 – July 2023)

- Designed the whole Quadcopter, Designed the PCB, the 3d printed frame, the Flight control software, and Flutter App to control it via phone.
- Ordered all components, assembled, debugged the Quadcopter, and flew it **[Video]**. Wrote a **Documentation website** using Read The Docs.
- Flight control software was written in Arduino and compiled using PicoW core. Wireless communication is done over WiFi via UDP protocol.
- Many articles were written on this Project in various languages English, French, Finnish, and Polish [**English article from TOM's Hardware**].

3D Rubik's Cube virtual Model(OpenGL C++), Scanner(OpenCV Python) & Self Solver(Koceimba Python) [GitHub] (May 2022 – May 2022)

- Built the virtual Rubik's cube 3d model in C++ using the OpenGL graphics API. Total code written was about 1500 lines.
- Implemented Rotation operations for Scrambling the cube, Lighting, Material Properties of the cube's faces and self solving via Kociemba library.
- Tested and manually assigned HSV colour ranges required for scanning colours of a real cube via webcam using OpenCV Python.

Skills

Computer Skills:	C, C++, Python, Java, Arduino/AVR, Linux, Git, GitHub, Perforce, Flutter, React, Latex, Keras, ROS, MIPS, VHDL, Verilog, Simpy, Pandas, Numpy, gdb, Vim, Django, Read The Docs, Bash, Haskel, OpenCV, OpenGL, Unity, IOT.
Other Skills:	PCB design, AutoCAD, Video Editing, Robotics & building RC plane/Quadcopters, MS office, Blogging, YouTube, Canva.
Relevant Coursework	Data Structures and Algorithms, Computer Architecture, Computer Graphics, Machine Learning, Artificial Intelligence, Operating Systems, Digital System Design, Computer Networks, Compiler Design, Cryptology and Coding Theory.

Positions of Responsibility

Student Representative 2020:	Represented Batch of 2020 in front of the UGC (Undergraduate Committee) of IIT Goa	(2020 – 2024)
Head, E&R (Robotics) Club:	Built line follower Robots, and RC planes. Started and Lead the Quadcopter R&D project	(2022 – 2023)
Core member of Infosec Club:	Organised and Competed in local or international Cyber Security events (CTFs).	(2020 – 2022)
Core member of Google DSC:	Organised HackTheGames event in Cepheus and was the Facilitator for Google Flutter Festival	(2021 – 2022)
Facilitator for Flutter Festival:	Conducted five session on Dart, Flutter and building apps using Flutter Live for DSC IIT Goa.	(2022 – 2022)
Volunteer, ACM Goa Chapter:	Edited event recordings and managed their **YouTube channel**, and WordPress website	(2021 – 2024)

Achievements

- Placed 3rd on the IIT Goa IIC virtual lab hackathon and won a cash prize of Rs 3000. **(Certificate)** (2021)
- Placed 1st in the IIT Goa's Coding contest, Game of Codes, and won a cash prize of Rs 300. (2023)
- Became a Microsoft Certified Technology Associate Python developer in first year. **(Certificate)** (2021)